

Manual

Graphic Planning Board HLS-GPT 8.2

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1 Overview: Graphic Planning Board

Purpose

This function package provides basic features of the graphic planning board. The graphic planning board clearly shows planned operations, specific machines and scheduled times. Using the intuitive operating concept of the graphic planning board you can quickly and efficiently plan, replan and deallocate operations.

You use the function package if:

- you want to plan operations in a graphical user interface
- you want to get an insight into current planning
- you want to quickly identify and use free capacities in production

Features

- Time frames, zooming, scrolling
 - Planning board with variable time frames, zoom and scrolling functions, shift calendars as basis for the available capacity
- Gantt chart
 - Gantt chart with operations (machine assignment)
- Graphic order backlog
 - Graphical presentation of the order backlog for machines / workplaces or machine / workplace groups
- Tabular order backlog
 - Tabular presentation of the order backlog for machine or workplace groups
- Tabular machine assignment
 - Tabular presentation of planned operations (machine assignment/occupancy) for each machine / workplace
- Color-coded OP bars
 - Color-coded display of setup times, startup times, production times, retooling times and idle times or of various operation statuses (active, interrupted, done). Coloring can be defined individually.
- Tooltip
 - Configurable info window showing detailed information (tooltip) about operations
- Display of planning conflicts and schedule violations
 - Display of planning conflicts (double occupancy) and scheduling violations (delays)
- Display of current BDE uploads
 - Current BDE uploads (machine statuses, operation statuses) are shown

- Conflict list
 - Conflict list with detailed information about planning conflicts (e.g. delayed operations, resource not available, resource overloaded, etc.)
- Print function
 - Print functions with page preview and various tools
- Ability to navigate to other applications
 - Ability to navigate in workplace overview, order information or order overview

Graphic planning board

Overview

	→	
HYDRA menu	Production control → Preparations for production → Graphic planning	
FEDRA menu	Detailed scheduling → Planning → Graphic planning	
Transaction code	grap	
Function authorization	grap	

HYDRA menu	Production control → Preparations for production → Info shop floor planning	
Transaction code	grapi	
Function authorization	grapi	

Available user fields		
Where?	Object type/user field key	Source (type)
Tabular <i>Pool of groups</i>	AUNR/SYSTEM	Order (MF-D)
Tabular <i>Pool of groups</i>	AGNR/SYSTEM	Operation (MF-D)
Tabular <i>Pool of workplaces</i>	AUNR/SYSTEM	Order (MF-D)
Tabular <i>Pool of workplaces</i>	AGNR/SYSTEM	Operation (MF-D)
Tooltip	AUNR/SYSTEM	Order (MF-D)
Tooltip	AGNR/SYSTEM	Operation (MF-D)
How to configure user fields? Which user field types are available?		

Purpose

You use the graphic planning board if:

- You would like to have the updated planning situation displayed,
- and/ or
- You would like to make planning changes.

Integration

You can use the graphic planning board to plan capacities for machines and workplaces in your production department.

If you apply the graphic planning board as information tool, you only use it to obtain an overview about latest planning; no changes can be made in the info shop floor scheduling application.

Requirements

Proceed as described below in order to display the most up-to-date planning in the graphic planning board:

- You have created orders that are relevant for planning in the system (production orders, maintenance orders, project orders) or you have downloaded these orders from a higher-level system.
- You have coordinated responsibilities within your company as concerns order planning and you have defined which workplace/machine groups should be planned. Based on these initial steps, you have defined the necessary [Planning profiles](#).

In order to [Change the planning](#) in the graphic planning board, you must:

- start the graphic planning board via the menu item *Graphic planning*, and
- have been issued the corresponding planning function authorizations.



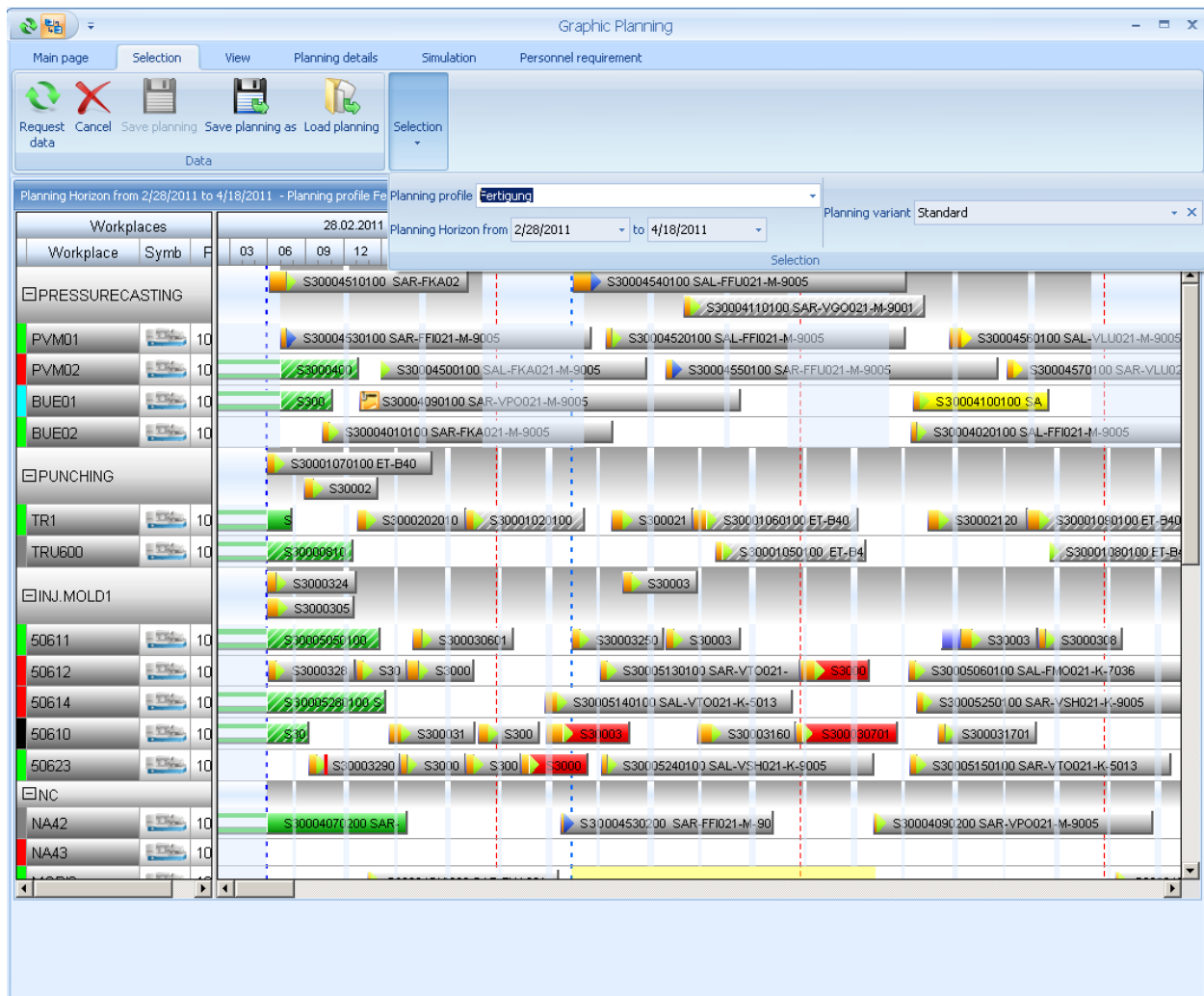
Please note that "workplace assignment" of the Personnel Scheduling module (PEP) and the graphic planning board should not be used simultaneously. Inconsistencies might occur if personnel and capacities are planned at the same time, as the "workplace assignment" application uses data structures of "graphic planning" (only applicable if HYDRA is used).

1.1 The graphic planning board

The graphic planning board is the basic element of the planning system. The graphic planning board is modeled after a traditional planning board. The planning board compares capacity groups including individual capacities (workplaces/ machines) with the pool of orders (pool of groups) or the operations already planned for individual capacities.

All order types are relevant for planning that have been configured using the ID "Planning". You can customize these [Order types](#) in the system.

The following chart illustrates the "Graphic planning" dialog:



Please note: This screenshot illustrates *one* possible presentation of the planning board. Depending on the settings, the presentation may vary.

1.1.1 Toolbar

This chapter describes the functions that you can call via the toolbar. Please keep in mind that the ability to use certain functions depends both on which licenses are available as well as how the function authorizations were assigned.

Furthermore, some functions are generally not available in "info shop floor scheduling" mode.

1.1.1.1 Tab Main page

Category Data

Data is requested

Data matching the selection criteria entered in the "selections" tab is read from the database and loaded into the planning board.

Each operation that is planned in the detailed graphic planning requires a remaining time formula. You use the remaining time formula to identify the planning duration of the operation.

HYDRA: as of service pack 16



FEDRA: as of version 1.1.

When requesting data, a note appears when operations are loaded where the remaining run time is missing.

HYDRA: as of service pack 16

Fedra: as of version 1.1



Starting with service pack 16, a note appears when operations are loaded with negative yield.

Cancel

Interrupts the process of requesting data.

Print all

Shows the entire print area. You can restrict or zoom the print area in this view. Refer to the chapter 1.1.8 *Print preview* for more information.

Save planning

Function authorization: grapt.save

The current assignment is saved. Changes are stored in the database and are thus available for other planners.

The internal fields used for sorting of the [Terminal's sequencing list](#) are set to the start date and start time as specified by the planner in the graphic planning board.

A pop-up dialog appears to notify the planner that changes have been saved.



A maximum of 32.768 operations may be changed and stored for planning purposes.

Lock all



The button "Lock all" is only available if the modification [graplocking](#) is enabled.

By clicking the button "lock all", you can lock all workplaces and groups of the planning scenario at once. Further information on the locking mechanism you can find [here](#).



This function does not require a function authorization.

Category *Planning function*

Automatic assignment

Function authorization: `grapt.autopl`

The function for [Automatic assignment](#) is called up. Doing so, this function distributes all of the operations to be planned in the planning board to the workplaces (capacities) based on the current allocation strategy.

Plan operation



The button "plan operation" is only available if the extension [grapt.sbsp](#) is enabled.

Function authorization: `grapt.sbsp`

[One-step planning](#) is carried out. This method plans all unplanned operations separately (step by step) according to the current planning algorithm.

Close gaps

Function authorization: `grapt.closegaps`

If there are any gaps between individual operations, the function [Close gaps](#) closes these gaps.

Generate campaign

Function authorization: `grapt.cpnbuild`

The function for [Generate campaign](#) is called up. Operations meeting specific criteria can be combined in a campaign.

Category *Options*

Search operation

All operations matching the criteria entered are selected.

Notes on the individual fields:

- Operation: this is a combined order/ operation number (MES order number).
- Article: The article/item stored in the operation (see *Order information* > *Operations* > tab *Operation* > field *Article*)
- Tool: The (main) tool stored in the operation (see *Order information* > *Operations* > tab *Operation* > field *Tool*)

You can also run a search using wildcards (at the beginning and/or end of a character string).

Next time you call the function, the dialog shows the selection criteria that you have entered the last time.

Settings

Here, you can modify different settings, such as changing bar colors. [Settings](#) describe the individual options.

Save

You can change the default layout of the application. You can e.g. change the following:

- Add or remove a detail application from the [Planning details](#) tab.
- Add or remove columns in a table and exchange the alignment of categories and columns in a table.
- Change the width of the groups and workplaces/machines displayed on the left hand side of the Gantt chart.
- Create settings in the dialog [Settings](#).

You must save these changes, if the changed dialog layout should still be available after having closed the application. Click the Save icon in the toolbar to save the changed application layout.

1.1.1.2 Tab Selection

Category Data

The functions are similar to the ones in Tab *Main page*.

Category Selection

planning profile

Select a [Planning profile](#) before requesting data. You use this planning profile to load, display and plan assigned capacity groups.

We recommend that planners with many responsibility areas make use of planning profiles. Benefits:

- Better overview

- Faster load times when requesting data

If you do not select a planning profile in the graphic planning board, *all* workplaces are loaded, which are included in the responsibility areas the planner is authorized for.

Planning horizon

The user can manually modify the dates *Planning horizon from/ to* at any time. The data selection always refers to the dates entered in the input fields. The system selects all operations with a scheduled start time within this period.

planning variant

Function authorization: grapt.plvar

There is also the option to work with so-called [Planning variants](#). A planning variant combines specific settings that are relevant for the planning. For the most part, these are settings taken into account when using the [Automatic assignment](#) function. Please keep in mind that the extent to which you can create and edit planning variants/versions depends on the license.

Order type

If you select one or several order types, only those operations are loaded and displayed that belong to an order of the respective order type.



Using this selection option, you can for example select and plan capacity or simulation orders. The planning of the production orders is not affected in this case. Important: Always include all order types when you select production orders before you perform and save a real planning.

1.1.1.3 Tab View

Category *Planning board*

Expand/collapse all groups

Opens or closes groups

Zoom in

You modify the Gantt presentation in order to increase the degree of accuracy of the time scale. Alternately, you can also enlarge the presentation using the key combination Ctrl + + (numeric keypad) or using the mouse wheel: turn the mouse wheel while keeping the Ctrl key held down.

Zoom out

You modify the Gantt presentation in order to get a better overview. Alternately, you can also reduce the size of the presentation using the key combination Ctrl + - (numeric keypad) or using the mouse wheel: turn the mouse wheel while keeping the Ctrl key held down.

General view

The general view is an additional window showing the complete Gantt chart and the [Histogram](#) (if shown). One separate frame is displayed showing each chart and histogram section that the main window is currently displaying. If you move the frame with the mouse, the section displayed in the main window is moved accordingly when you release the mouse button. Similarly, you can zoom the frame. The display detail in the main window is then changed accordingly. And vice versa, size and position of the frame also change, if you zoom or scroll the section displayed in the main window.

Note:

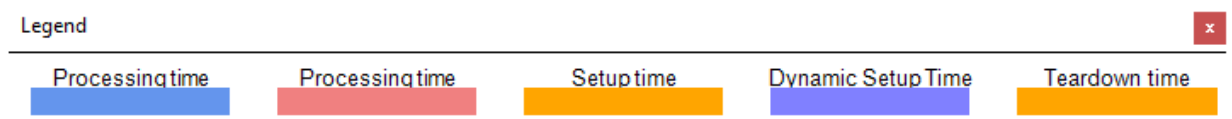
Open and close the *General view* by clicking the relevant button. Every time you request data, the *General view* is automatically closed.

Legend

The legend view is an additional window to present a legend on the screen. The following information is shown:

- Processing time (in some cases, shown multiple times because of different color codes!)
- Setup time
- Dynamic setup time
- Teardown/retooling time
- Operation fixed (blue arrow or red bar)

The configurations in the [Settings](#), section [Bar layout](#), specify the respective display.



Notes:

Open and close the legend by clicking the relevant button. Every time you request data, the legend is automatically closed.

Category Operations

Links

The relationships (links) between each of the operations are shown as lines.

The links are shown and removed by clicking the relevant button.

Priority

The priorities of each of the separate operations are shown as triangles. You can configure their color codings in the [Settings](#).

The priorities are shown and removed by clicking the relevant button.

Fixing

A symbol (triangle) or a red bar identifies fixed operations. You can configure their color codings in the [Settings](#).

The symbols or bars are shown and removed by clicking the relevant button.

Resource occupied/locked

You can identify reserved/blocked resources. The use of this function is described [here](#).

Notes on operations

This function displays the detail application *Notes on operations*. The use of this function is described [here](#).

Open and close the detail application by clicking the relevant button.

Show delays

You can identify delayed operations. The use of this function is described [here](#).

Category Adjust bar colors

The buttons of the category *Adjust bar colors* link to the respective functions in the HLS settings. For a description of the functions, please refer to the section [Tab Bar layout - Bar colors](#).

1.1.1.4 Tab *Planning details*

conflict list

Function authorization: `grapv.conflict`

A detail application is opened showing the planning conflicts for the current planning profile. For more information about the conflict list, refer to the details described [here](#).

Open and close the detail application by clicking the relevant button.

KPI

Function authorization: `grapv.keyf`

A detail application opens showing [KPI](#) to assess the current planning.

Open and close the detail application by clicking the relevant button.

Utilization profile

Function authorization: `grapv.capgroup`

A detail application opens displaying the [Utilization profile](#).

Open and close the detail application by clicking the relevant button.

Kapazitätsauslastung

Function authorization: `grapv.captotal`

A detail application opens showing the [Capacity utilization](#).

Open and close the detail application by clicking the relevant button.

Histogram

Function authorization: `grapv.hist`

The [Histogram](#) is displayed below the Gantt.

Open and close the detail application by clicking the relevant button.

Resource view

Function authorization: `grapv.resview`

The [Resource view](#) is displayed.

Open and close the detail application by clicking the relevant button.

Pool of groups

Function authorization: `grapv.tabgrp`

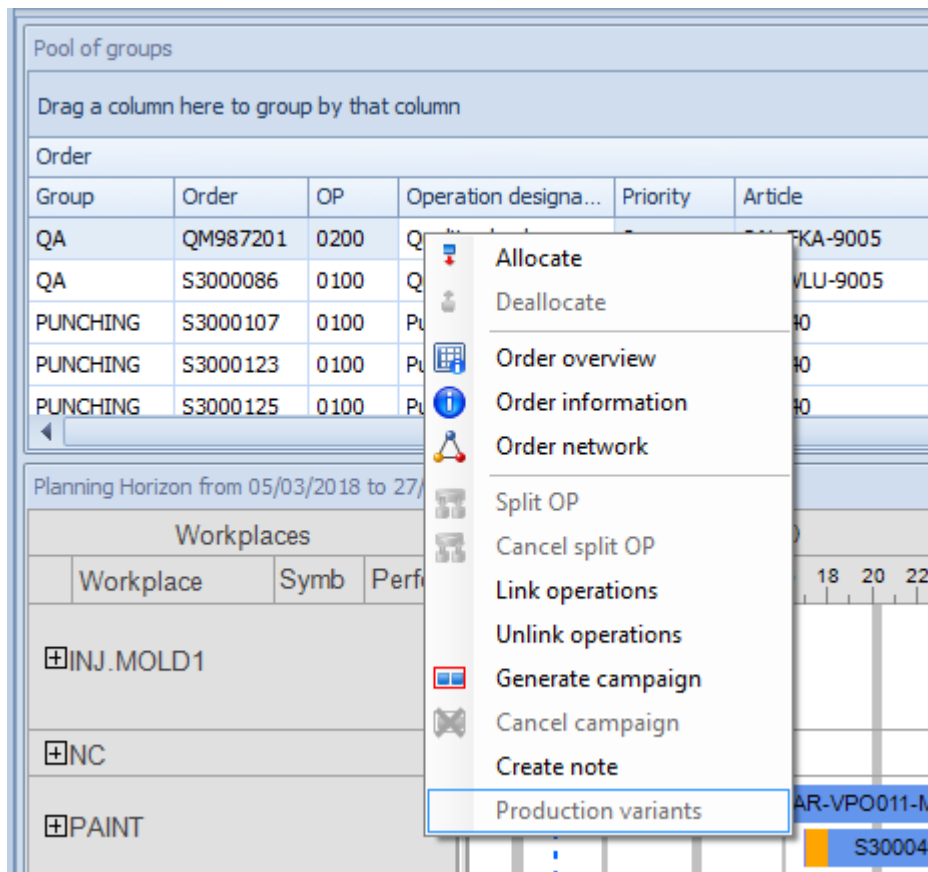
The function call *Pool of groups* opens a detail application showing the operations that are planned for a group. The display includes the scheduled operations for *all* groups.

Open and close the detail application by clicking the relevant button.

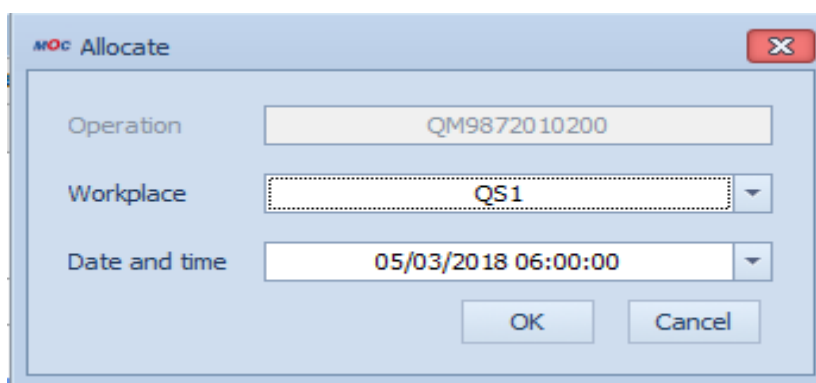
HYDRA: as of service pack 16

Fedra: as of version 1.1

Operations in the tabular and the graphic group list can be scheduled in the context menu.



You can select in the corresponding dialog the workplace/machine to be planned and the planned start (menu point "Point in time") for the operation.



Available workplaces/Workplace backlog

Function authorization: grapv.tabwp

The function call *Pool of workplaces* opens a detail application showing all scheduled operations of a workplace in tabular form.

Open and close the detail application by clicking the relevant button.

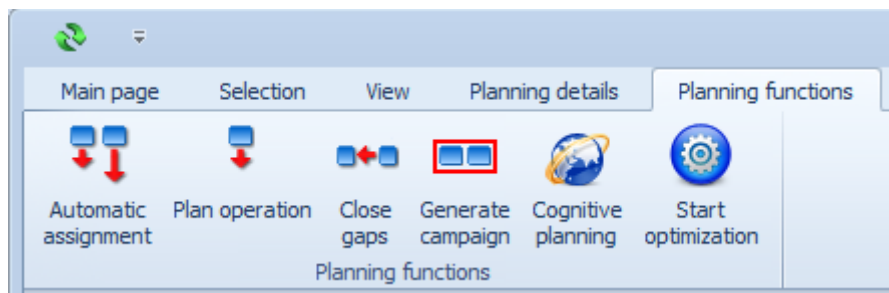
Order network

Function authorization: grapv.orview

An application opens showing the [Order network](#) the selected operation is assigned to.

Open the detail application by clicking the relevant button. Unlike the other detail applications, the order network is not closed by clicking the button again, but by closing the window.

1.1.1.5 Planning functions tab



HYDRA: as of service pack 16

FEDRA: as of service pack 1.1

There is a "Planning functions" tab as of Service Pack 16. The buttons for the following functions have been moved to this tab:

Automatic assignment

Plan operation

Close gaps

Generate campaign

Start optimization

The new tab also includes a button for the following function:

Cognitive planning

Reserved for future applications

.

1.1.1.6 Tab *Simulation*

The "Simulation mode" in the capacity planning table supports the creation and saving of different plans on the basis of initial situations and the comparison of different plans. The functions of the simulation mode are described at [here](#).

A description of the buttons included in this tab is provided [here](#)

1.1.1.7 Tab *Optimization*

The planner can use the "optimization" function to calculate several planning runs automatically. The system uses different parameters for the calculation of the individual planning runs and evaluates the planning runs using a KPI. The planning providing the best KPI during the optimization process will be used.

How the optimization process works is described [here](#).

1.1.1.8 Tab *Personnel requirement*

You can use the function *Personnel requirement* of the graphic planning to show and compare the required and available personnel in a histogram and a table.

The functions and input options provided in this tab are described [here](#).



The display of personnel requirements in tabular form is available as of Service Pack 15.

1.1.1.9 Tab *Energy consumption*

Function authorization: grapv.egfc

A detail application opens showing the energy consumption for operations. This function is only available, if you use the Energy Management EMG (only applicable if HYDRA is used).

Open and close the detail application by clicking the relevant button.

The graphic planning integrates seconds to display the planned energy consumption. We recommend to take this into account when you define the upper limit value.

Example (we used large numbers for demonstration purposes)



- An OP has a target quantity of 1,000 pieces, a processing time of 1:00 hour and an input quantity of 100 kWh that is stored as a component. In this example, the resulting required quantity of the OP is 100,000 kWh (target quantity x input quantity). In the graphic planning, an energy consumption of 27,78 kW is displayed for the OP (required quantity / processing time [sec]).
- In this example, the upper limit value is understood as load limit. If the supplier provides 27,000 kWh within a time frame of 15 minutes, an average of 30 kW is available per second in this time frame (provided quantity / time frame [sec]). Here, you should use 30 kW as upper limit value.

1.1.1.10 Tab *Planned inventory levels*

A detail application opens showing the material trend for materials or articles.

Open and close the detail application by clicking the relevant button. The functions and input options provided in this tab are described [here](#).

1.1.2 Time scale

A blue, vertical line identifies the current point in time "now". In addition, red vertical lines visualize the end of each day.

The dimension of the time scale is defined in the [Settings](#).

To change the resolution, resize the time scale (by holding down the left mouse button).

Another blue, vertical line shows the planning time fence. This time fence is specified in the planning variant. If it is 0, no planning time fence is used. If no planning variant is specified, the planning time fence of the basic settings is used. If it is 0, no planning time fence is used. If the planning horizon starts in the future ("From" > today), then no planning time fence is used.

Note

Every time you request data, the time scale is set back to its original setting.



The format of the date values displayed on the Gantt chart depends on the format specified by the operating system. The client format is not relevant.

1.1.3 Presentation of groups and workplaces/machines

The workplaces/machines that are available for planning are displayed on the left hand side. A workplace is shown if the following criteria are met:

- The ID "planning function" is set to 'P' or 'H' in the [Workplace and resource configuration](#).
- The workplace is not "blocked" in the *Workplace and resource configuration*.
- The workplace is assigned to a [Group](#) that is defined as a capacity group.
- The group that the workplace is assigned to, uses the selected [Planning profile](#). Here, the option *Visible in shop floor plan* is enabled.
- The workplace/machine is in the planner's [Responsibility area](#).

If you open the graphic planning board and request data, the overview of workplace groups is generally shown. The group number is shown. You can define the order of the groups using the field *Sequence* in the configuration of the [Planning profile](#).



Please keep in mind that the values in the field *Sequence* must be unique within a planning profile. Otherwise the order is random.

Select a group by clicking it. The workplaces/machines of this group are shown. Use the field *Position* in the *Group assignment* configuration to specify the order in which workplaces of a group are displayed. Please keep in mind that the values in the field *Position* must be unique within a group. Otherwise the order is random.

List of workplaces: Columns

The following columns are available for display in the list of workplaces:

Header	Meaning	Comment
(none)	Status color or RPA color (according to the Setting)	Due to the linked logic, group workplaces always have the status <i>Production</i> (assigned to the resource performance account <i>Main utilization</i>). You can define the update interval of the status in tab <i>General</i> of the settings. Note: If the option <i>Show personnel assignment</i> is enabled in the Settings , the symbol [+] or [-] is shown. This is the case for all workplaces, even though no staff is assigned in the Personnel Scheduling (PEP) (only applicable if HYDRA is used).
Workplace	Number of the workplace/machine	

Header	Meaning	Comment
Short name	Short name of the workplace/machine	
Designation	Full name of the workplace/machine.	
Group	Group of the workplace/machine.	
Symbol	Graphic	
Performance level	Current performance level of the workplace/machine.	The performance level is read from the database when data is requested. When updating subsequent statuses, the performance level is not updated.

The planner defines in the tab [Workplace](#) of the settings which columns are displayed in the table. The configuration is saved user-specifically. You can also change column widths by "dragging" columns in the table. This modification is also stored for the specific user.



If the extension [graplocking](#) is enabled, workplaces or opened workplace groups, which have been blocked as part of the planning process, are highlighted in color. You will find further information on the display of personnel capacities in the Graphic Planning [here](#).

Tooltip

If the mouse pointer rests on a workplace, a tooltip is displayed providing the following data:

Header	Meaning	Single workplace	Group workplace
Workplace	Number of the workplace/machine according to configuration	X	X
Short name	Short name of the workplace/machine according to configuration	X	X
Designation	Full name of the workplace/machine as configured	X	X
Cost center	Cost center of the workplace/machine as configured	X	X
Workplace type	Workplace type as configured: E = Individual workplace G = Group workplace	X	X
status	Status number and name of the workplace/machine status as configured	X	-
Status since	Point in time (date, time) since when the status has been active.	X	-

1.1.4 Functions on the level of the group/workplace

You can perform the functions described in the following if you select a workplace or group in the column *Workplace* and right-click it (to open the context menu). The order in the context menu may vary from the order the functions are described here:

Workplaces/Machines

This menu item starts the application [Workplaces/machines](#) for the current workplace/machine or group. The workplace number or group is transferred to the application as selection criterion.

Close gaps

Should there be any gaps between individual operations, you can use the [Function](#) to close these gaps.

If you open the context menu on group level, this function closes the gaps of all workplaces belonging to the selected group. If you start the context menu from a workplace, you can only close the gaps for this workplace.

Change shift model/ performance level

The entry is only available if you request the context menu from the workplace. You can change the following configurations for a workplace without having to switch to the workplace/machine configuration:

You can change the performance level of a workplace/machine. The performance level is used to calculate the processing time/[Remaining run time](#).

In addition, you can also define or change a year model used for planning that is not identical to the year model used for data collection.

Please keep in mind: this data is only stored if you save the planning.

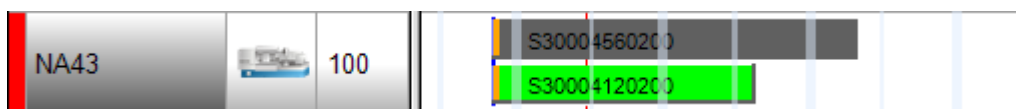
Add individual shift time

Call this function to define a time without shifts for a workplace that is different from the times defined in the shift model. You can find more information about individual shift time [here](#).

Single-line presentation

With single-line presentation, operations assigned twice are not displayed one below the other (i.e. listed) but one on top of the other (i.e. overlapping).

Two-line presentation:



Single-line presentation:



Note:

- With single-line presentation the text on OP bars might partly not be visible
- It is only a temporary presentation option. Every time you request data, parallel operations are again displayed one below the other.

Select operations in the pool of groups

This entry is available if you call the context menu on group level. Click this entry to select all operations included in the pool of groups.

Select planned operations of the group

This entry is available if you call the context menu on group level. Click this entry to select all operations planned on the workplaces of the group.

Select planned operations

The entry is available if you request the context menu on workplace level. Click this entry to select all operations planned for this workplace. Also fixed operations are selected.

With the above-mentioned functions and the functions *Select all operations of the order* and *Select this OP and all following OPs of the order* that are included in the operation's context menu, you can select multiple operations by holding down the CTRL key, i.e. the selected amount of operations remains and the newly selected operations are added.



If you want to select all operations planned for workplaces in the current planning scenario, select the function *Select planned operations* from the operation's context menu.



The functions *Select all operations of the order* and *Select this OP and all following OPs of the order* that are included in an operation's context menu only affect planned operations.

1.1.5 Displaying operations in the Gantt

Order backlog / pool of groups / pool of orders

Operations transferred from the ERP system are generally planned for groups and are therefore located in the pool of groups. You can show the pool of groups in [tabular form](#) and/or as Gantt chart.



If the machine/capacity group is changed for a workplace, you must **also** change the group for the planned operations in the order backlog. Otherwise, the planning board no longer shows the operations.

Pool of groups in Gantt

In order to graphically display operations in the pool of groups of the Gantt, you must activate the option *Show operations in pool of groups of the Gantt* in the settings in tab [General](#). In this case, operations are displayed according to their scheduled start time.

For the pool of groups, the system shows the times without shift of the shift model that is defined for the group.

Planned operations

Planned operations are displayed for the individual capacity (workplace/machine), i.e. in the relevant row (only if the view is expanded).



If you log on an operation to several workplaces/machines, the graphic planning board will show this operation as logged on to the workplace/machine where the operation has last been logged on to. If you want to show the operation in the graphic planning board with several workplaces/machines, you must [split](#) the operation.

If you log off an operation from a workplace, the display of the workplace does not change in the graphic planning board. The same applies if the operation is no longer logged on to workplace scheduled last.

Duration/length of an operation bar

The length of an operation is the sum total of the setup time, processing time/[Remaining run time](#) and retooling time (teardown time). This time is called execution time. You can define the process times either directly or the system can calculate these times using a formula.



The planning board generally includes the setup time. The setup time is also included if the processing of the operation has already started or if the status "interrupted" is available at the moment.

But you can customize the system and configure if you want to integrate the setup time for interrupted operations in the planning board. To configure the order type, the following other options are available :

- Integrate remaining setup time: target setup time minus "Times posted to RPA 7"
- Ignore setup time if processing of operation has already started (even if OP is interrupted)

Running operations do generally not include the setup time.



If the current remaining run time is 0 or negative for an operation (e.g. in case of overproduction), the remaining run time is set to 0 for a running operation. In case of a prepared or interrupted operation, the remaining run time is set to 900 seconds = 15 minutes by default. This way, this operation can still be re-planned manually using the drag & drop function in the Gantt. You can change this duration planning board settings in tab [Bar layout](#).

Color coding of operations

You can define the color of operation bars in the settings in tab [Bar layout](#).

Exceeding basic dates

If an operation is planned outside the predefined basic dates and the planned end date is later than its "latest end date" (LET), the presentation changes. The operation bar is then hatched or displayed in a different color. This requires that the option "Operation delayed" is enabled in the settings, tab [Bar layout](#).

Double clicking an operation bar

Double clicking an operation not logged on in the Gantt will open the dialog to modify default values/target data. To edit the data, you require the authorization *grapt.edop.edit*. You can change the following data in the dialog:

- Target quantity (primary quantity unit)
- Partitioning
- Setup time
- Target cycle
- Teardown/retooling time
- Wait time

Confirm the dialog to store the changed data directly in the operation and to update the planning board.

1.1.6 Operation functions

If you right click a selected operation, a pop-up menu opens offering the following functions:

Fix operation

If you select this option, you can [fix](#) one or several operations. The settings (see tab [Bar layout](#)) specify how fixed operations are visualized.

Unfix operation

If you select this option, you can [unfix](#) one or several operations.

Deallocate

Use this function to deallocate selected operations. The operations are then available in the pool of groups. For more information, see [here](#).

This menu entry is not available for fixed operations.

Order overview

The [Order overview](#) is opened for the selected operation. The order number of the selected operation is transferred as selection.

Order information

The [Order information](#) with relevant default data and current information about the order is opened for the selected operation.

Order network

This menu item opens the application [Order network](#) for the selected operation. The order network shows the operations that are related to the selected operation in a graphic (Gantt chart).

This function requires the respective authorization (license).

Split OP

Subject to the setting of the option *Enhanced split function* in the [Basic settings](#), you can find the requirements and more information about operation splitting in one of the below-mentioned documents:

- Enhanced split function is not enabled: [Operation Split](#)
- Enhanced split function is enabled: [Enhanced split function](#)

This function requires the respective authorization (license).

Cancel OP split

Use the function *Cancel split* to undo the splitting of an operation.

Subject to the setting of the option *Enhanced split function* in the [Basic settings](#), you can find the requirements and more information about operation splitting in one of the below-mentioned documents:

- Enhanced split function is not enabled: [Operation Split](#)
- Enhanced split function is enabled: [Enhanced split function](#)

Generate campaign

You can combine OPS to a campaign. You can find further information on campaign production at [This document](#).

Cancel campaign

You can also cancel campaigns. You can find further information on campaign production at [This document](#).

Join operations

You can join operations. Further information you can find on joint production at [This document](#).

Cancel joined operations

You can also cancel operations you have joined. Further information you can find on joint production at [This document](#).

Create note

You can create one or more notes for a selected operation. In order to display notes, you must enable the corresponding view [Notes](#).

Production variants

You can select an alternative [Production variant \(method\)](#) here if, for example, the tool is not available that is needed for the planned variant. From this overview, you can switch to the editing mode and edit the production variants if required.

This function requires the respective authorization (license).

Select all operations of the order

This function selects all operations included in the Gantt chart and belonging to the same order as the current operation.

In connection with the function *Deallocate*, you can remove an entire order from planning. Please note that you first have to unfix fixed operations.

Select this OP and all following OPs of the order

Use this function to select this operation and all operations included in the Gantt chart that belong to the same order and have a higher operation number than the current operation.

Select the planned operations

This function selects all operations planned in the Gantt chart.

In connection with the function *Deallocate*, you can remove all operations from planning. Please note that you first have to unfix fixed operations.

1.1.7 Presentation of relationships / network view

The Gantt chart shows the relationships between operations. You can choose from the following options:

- Click one operation to show the operation's relationships.



- Click the button in the toolbar to visualize all relationships and dependencies of all operations. A complex presentation then shows all order relationships at the displayed workplaces.

Please note: Only the relationships relevant for planning are displayed.

1.1.8 Print preview/Print



The functions and options described in this chapter are part of the used Gantt visualization component. As this is a standard third-party product, changes to the layout and the functionality are only possible to a limited extent.

You can print the planning board using the print preview screen, which can be opened by clicking the



button in the tab [Different screens](#) of the toolbar.

You can view each single page or see an overview of all pages of your presentation. You can also interactively zoom in on a section of your chart and then print it.

The status line provides information about the total number of pages and the horizontal and vertical distribution of pages. In the view *Show single page*, the current page number is additionally shown.

Functions of the print preview

Close

You exit the print preview and return to the application.



Note: The settings made in the print preview are currently not saved.

<

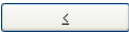
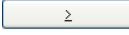
Only active if the button *Show single page* has been clicked. If the chart takes up several pages, you can view each single page. Click this button to return to the previous page. You move through the pages from right to left in ascending rows.

>

Only active if the button *Show single page* has been clicked. If the chart takes up several pages, you can view each single page. Click this button to go on to the next page. You move through the pages from left to right in descending rows.

Overview

If the chart takes up several pages, you can either view each single page separately or in the overview. In the *Overview*, all pages are displayed - depending on the number of pages, the view is zoomed out more or less; in the *Single page* presentation mode, initially the first page of the chart is shown and zoomed in.

Click  or  to scroll through the pages. Double-click a page to easily switch between the two presentation types *Single page* and *Overview*.

In the presentation mode *Single page*, you can interactively enlarge specific sections of your chart. To do so, hold down the left mouse button and make a square frame around the section you want to zoom in. As soon as you release the left mouse button, the framed-in image section is enlarged. The label text of the button *Print* changes to *Print area*. You can now print the image section in the currently enlarged size. Note: The zoom factor selected in the print preview does not change the scaling factor in the dialog field *Page setup*.

Fit to single page

Use this button to reduce a chart of several pages to one page. Here, you also have the possibility to interactively enlarge sections of the chart as described in *Single page/ overview* and to print them.

Zoom factor

You can use a zoom factor from the list to change the presentation size of your chart in the print preview. You can only select the zoom factor if single pages are shown. The print is not affected. Depending on the selected size, vertical and/or horizontal scroll bars are displayed. You can also use the mouse wheel to move the image (without shift key vertically, with shift key horizontally).

Auto is preset as zoom factor. With this setting, the size of the page is always reduced or enlarged so that it fills the screen.

Page setup

If you click this button, you are forwarded to the dialog *Page setup*, where you can change the page layout. The options of this dialog are described below.

Print/print area

If you click this button, you are directed to the Windows dialog *Print*. If you have zoomed in a section in the print preview, the label text of the button changes to *Print area*. If you click this button, the option *Location* is preselected in the Windows dialog *Print*. Click *OK* to print the section shown on the screen.

Note: The zoom factor selected in the print preview does not change the scaling factor in the dialog field *Page setup*.

Functions of the dialog *Page setup*

Mode

Select the scaling type from the drop down list and the relevant values for *Zoom factor* or *Maximum width/height* to define the scale of the presentation's output. If you click *Apply* , the values resulting from your settings are displayed in the section *Current*.

Zoom factor

If the scaling factor is 100%, the original size is shown. A smaller value will reduce the presentation, a larger value will increase it.

Fit to page count (Adjust to the number of pages)

Select this option to define the maximum number of pages for the width and the height. The chart output is then divided accordingly (**maximum width, maximum height**). The charts are presented as large as possible, but they are not distorted.

Zoom with horizontal fit

Select a zoom factor and a fixed number of printed pages (width) that should be maintained by compressing or stretching the time scale during printing.

Repeat title/ table/ time scale/ legend

Check this option to include title, table, time scale and legend on each page of the printed chart.

Show table

Define whether or not to print the table. If this option is not selected, the table is not printed.

Table columns

Here, you define which table columns are printed. Enter the required columns separately or in ranges separated by a comma or semicolon. Example: "1;5-7;3" selects the columns 1, 3 and 5 to 7.

Show diagram

Here, you enable or disable the printing of the chart (time scale and bars).

Adjust time scale to width of pages

This option optimizes the space used on the printed pages:

- If a fixed number of pages is defined for scaling: The zoom factor is calculated so that the full height of the set number of pages is printed. At the same time, the time scale is compressed or stretched so that the full width of the defined number of pages is made use of.

- If a zoom factor is defined for scaling: The time scale is compressed or stretched so that the full width of the defined number of pages is made use of.

Pad pages with space

Use this option to define whether you want to leave space between the chart and the boxes for title and legend. If you enable this option, the boxes are always positioned on the page margin and printed in full width on each page. If this option is disabled, the boxes are printed without space next to the chart. Depending on the chart, the width of the boxes may vary on the different printed pages.

Show frame outside

Activate this checkbox so that a frame is printed around the chart. If the option *Repeat title/ table/ time scale* is enabled, a frame will be printed on each page; otherwise a frame will be drawn around the entire chart.

Alignment

Define the alignment of the charts on the page. The printer settings define the page orientation (portrait or landscape).

Show crop marks

If you enable this option, crop marks are set on the chart. This way it is easier to glue the printed single pages to create a full chart.

Show folding marks (DIN 824)

For construction diagrams, the German national standard DIN 824 defines a very specific folding procedure that allows you to fold up the drawing to a DIN A4 format size. If this option is enabled, the folding marks on your diagram facilitate the folding procedure. The following formats are available:

- **Form A:** With a binding margin on the left side so that holes can be punched in the folded drawing and can be placed in a binder without needing binding strips.
- **Form B:** All in all, the format is somewhat narrower so that a binding strip can be attached, and combined with the drawing, achieves a DIN A4 format width.
- **Form C:** No holes are punched onto the folded drawing, but instead the drawing is placed in a transparent cover.

The folding marks can be added for any target format, whereas DIN 824 only includes the formats DIN A0 to A3.

Page numbering

If you check this option, the page number is printed on the bottom left of every page. You can select from the following options:

- **Row.column:** This option makes sense if the chart extends beyond the page length or even its width. The position of the page in vertical order is printed before the dot, the position in horizontal order is printed after the dot.
- **Column.row:** This option makes sense if the chart extends beyond the page length or even its width. The position of the page in horizontal order is printed before the dot, the position in vertical order is printed after the dot.
- **Page/ total number:** The first number shown is the current page number followed by the total number of pages: 1/6, 2/6 etc.

Text

Enable this option, if you want to add a random text at the bottom left of every page. In some cases, this text is printed to the right of the page number.

For page numbering, you can enter the following placeholders in the line **Additional text**. These placeholders are then replaced with the relevant content when the document is printed.

{PAGE} = Consecutive page number

{NUMPAGES} = Total number of pages

{ROW} = Row position of the detail section in the total chart

{COLUMN} = Column position of the detail section in the total chart

Print date

If you check this option, the print date is printed on the bottom left of every page. In some cases, the print date is printed to the right of the page number and the additional text.

Sheet margins

With the fields **Top**, **Bottom**, **Left** and **Right** you can define in cm the space between the paper's margin and the chart. You cannot go below the minimum margins that printers preset for technical reasons.

1.2 Settings



If you click , you call a dialog where you can make user-specific settings. The settings are saved user-specifically. The next time you call the graphic planning, the settings are available.



Some changes are only activated, when you request data the next time. For this reason, we recommend refreshing the data within the planning board, once changes have been made.

1.2.1 Tab *Main page*

General settings

Update statuses

Use this setting to define if and at which interval the workplace/machine status is read. The updated status is displayed on the left hand side of the planning board (column without caption).

Update data

This setting defines if and at what interval the planning board is updated (i.e. data is read and displayed). If changes have been made in the meantime, a prompt pops up asking whether the changed data should be saved or not.

Update planning scenario automatically while requesting data

If this option is set, then corrections are made to the planning scenario when the planning board is updated: any delayed operations (operations that were planned in the past and which have not yet been processed or started) are planned in the order in which they were originally planned. The relative gap between an OP plan start and the current line remains. If this results in a conflict (resource conflict, order network conflict) with operations planned in the future, they will be postponed further (future).

Note: This function only takes the assignment of the workplace into account. Fixed operations are also moved.



This function can only be used if none of the workplaces/machines shown in the planning table contains more than 350 operations.

Update order overview automatically when selecting an operation



The function is available as of Service Pack 16.

If you click on one of the following elements, the order number of the operation is copied into the [Order overview](#) and the display in the order overview is updated:

- Operation bar in the planning board (Gantt)
- Operation in the tabular pool of workplaces
- Operation in the tabular pool of groups

Following, you see the order progress for the whole order in the order overview.

Requirement: The option "Update order overview automatically when selecting an operation" is set and the order overview is opened in parallel.



Each click updates the order overview and switches it to the front. Therefore the order overview should be placed on a second screen.

If you click on different operations too quickly in succession, the update may be delayed and a popup will be displayed, indicating that data queries are still running and asking you to wait for the result.

Presentation

Period of the past

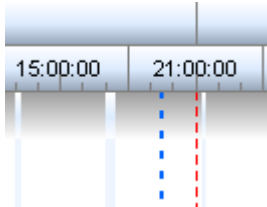
This setting specifies the range/period that is displayed left to the blue indicating "now".

Time scale

Define the time scale that you want to show on top of the chart in the planning board. Possible values: Minutes, hours, days, weeks, months.

Display planning time fence

If this option is enabled, a vertical blue line visualizes the planning time fence.



The planning time fence is specified in the planning variant. If it is 0, no planning time fence is used. If no planning variant is available, the planning time fence of the basic settings is used. If it is 0, no planning time fence is used. If the planning horizon starts in the future ("From" > today), then no planning time fence is used.

Show operations in pool of groups of the Gantt

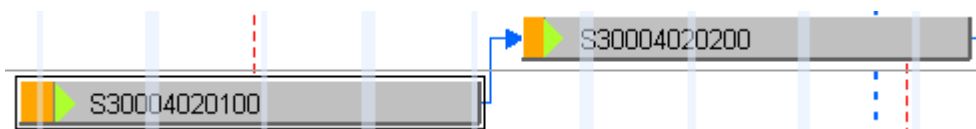
If this option is enabled, the operations included in the pool of groups are shown in the pool of groups of the planning board (Gantt). The operations are shown there at their scheduled date.

If you want to display the operations only in the Gantt pool of groups, you must hide the tabular pool of groups.

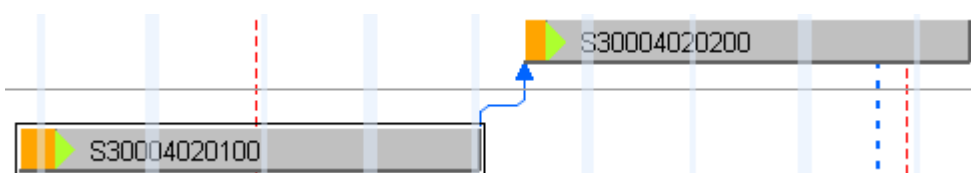
Please note: One of the two presentations of the pool of groups should be visible at all times.

Alternative presentation of relationships

By default, the lines between the operations are shown in the following manner:



With the option "Alternative presentation of relationships" the lines between operations are shown in the following manner:



Note: If you select this presentation, the size of each row may change dynamically.

Save options

Fix operations within planning time fence when saving

Operations, which are scheduled in the graphic planning board and have a planned start date within the planning time fence, are automatically fixed when the planning board is saved. This way, these operations cannot be re-planned.

Save automatically updated planned dates

When you exit the planning board or when new data is loaded, a prompt asks if you want to save the changes. This request is different in case of changes that are the result of explicit planning changes (manual planning, start of the automatic assignment,...) or in case of changes that are due to the automatic update (moving delayed operations (planned and running) to the "Now line").

- The option "**For planned operations**" is set:

The prompt that asks if you want to save the planning also appears if operations are implicitly re-planned.

- The option "**For running operations**" is set:

Running OPs are saved if the planned end date has been recalculated.

1.2.2 Tab *Planning component*

You can enable or disable the following validation checks for interactive user action:

Checks with interactive planning



Further information on interactive planning can be found in the document entitled [Detailed planning and assignment functions](#).

The relevant checks are processed in sequence. If a user action (planning/re-planning) is not allowed following one of the (enabled) consistency checks described below, then the action is canceled, and the remaining checks are not performed. The conflict is displayed in a pop-up window, where the planner can decide which steps can be taken to resolve the conflict:

Check basic dates of operation

The system checks if the start of the operation is within the operation's time frame when you schedule the operation for a workplace or when you move the operation within the time frame [earliest start, latest end]. The operation must be entirely between the earliest start and the latest end.

Note: The dates earliest start time (EST), latest start time (LST), earliest end time (EET) and latest end time (LET) must either be downloaded from the ERP system or can be calculated in the MES using the [lead time scheduling](#).

Check for relationships

The system checks if the operation maintains its relationships to the preceding and subsequent operations that are defined in the order network when you plan the operation on a workplace or when you move the operation to a start time X.



The application can only check preceding and subsequent operations if they are included in the current planning profile.

Check for capacity availability

When you move an operation and schedule it at a specified point in time, the system checks if the resources that are required to produce the operation are overloaded.

Here, "resources" refers to the workplace as a primary resource (workplace on which the OP is planned or re-planned) and the resources assigned to the operation (production resources and tools). However, the production resources and tools can only be taken into account if they are defined in the system and if their resource type is specified as "relevant for planning".

If a validation check is deactivated, resulting conflicts are not displayed, i.e. a possible conflict is implicitly confirmed.

Warning if operation is re-planned onto another group

If operations are planned onto another group than the one intended in the graphic planning board, a confirmation prompt is issued as warning.

Check material availability



This option is only available if you enable the extension [grapvemvp](#).

The system checks all materials required by the operation and having a valid [ATP inspection group assignment](#). A conflict message is shown if required material is not available. You can find further information on the material availability check [here](#).



Material availability is only checked if the detail view [planned inventory levels](#) is shown. In manual planning, the system informs you of potential material conflicts during the planning activity. You can view the material conflicts that exist in planning in the conflict list.

Check personnel availability



You can enable the check for personnel availability in relation to workplaces in the [Workplace and resource configuration](#), tab "workplace configuration" in the "HLS" section. The system only carries out the check if the option "show personnel assignment" is also enabled in the

"workplace" tab.

Processing with automatic allocation

Fix operations in planning time fence after automatic assignment

If the option *Fix operations in planning time fence after automatic assignment* is set, the system automatically fixes the operations if their planned start time is within the planning time fence after the automatic assignment.

1.2.3 Tab *Bar layout*

Bar colors

Color gradient

If this option is set, operation bars are displayed with a color gradient in the planning board .

3D effect

(reserved)

You can choose from the following options to display the processing time in color:

status

The color defined for the respective operation status is used. You define the color in the status assignment. This way, currently logged on OPs can be identified via the respective color.

Predecessor status

The color is used that has been defined for the status of the predecessor operation in the status assignment.

Order

Operations with the same order number are shown in the same color. The color is selected randomly using internal 8-color iteration.

Order type

Operations with the same order type are shown in the same color. The color is selected randomly using internal 8-color iteration.

Article

Operations processing the same article are displayed in the same color. The color is selected randomly using internal 8-color iteration.

Tool

Operations using the same tool are displayed in the same color. The color is selected randomly using internal 8-color iteration.

Processing time

The bar of the processing time is displayed in the color that has been configured.

Color from OP user field

If a color has been defined for an operation in the configured user field (possible user fields: 7 - 22), the operation is displayed in the respective color. In this case, coloring depends on the number entered in the user field.

Number	Color
1	Green
2	Bright red
3	Pink
4	Purple
5	Black
6	Gray
7	Turquoise
8	Light blue
9	Blue
10	Brown
11	Light green
12	Yellow
13	White
14	Olive
15	Silver
16	Light yellow

Color of running OPs

Running operations can be displayed in a user-defined color.

Display setup times

Activate this option to display setup times as an integral part of the operation bar. If you have enabled the option, you can display setup time, dynamic setup time or retooling time (teardown time) in one of the available colors.

Dynamic setup time

Select the color in which additional setup times should be shown. Dynamic setup times require a respective configuration in the setup time matrix (depending on license).

Setup time

Then, select the color for the display of the setup time that is stored in the operation (fixed).

Planned teardown time

Select the color for the display of the teardown time that is stored in the operation.

Other visualization

Operation fixed

If this option is set, fixed operations are either shown with a symbol (blue arrow with the tip showing downward) in the center of the processing bar or with a vertical red bar at the beginning of the processing time.

Note: You can also use the toolbar to enable/disable the symbol used to display fixed operations.

Operation delayed

If this option is set, delayed operations are shown with a hatched pattern or in color. An operation is considered delayed if its planned end extends beyond the latest end date (LET/SEZ).

Resource occupied/locked

If the Tools and Resources Management (WRM) is active, the system additionally loads the resource allocation when orders are loaded into the planning board. It is also checked for double allocations and locked resources. If the resource is already occupied by another operation, or if the resource is locked, the bar is shown in a different color.

If you manually move operations, the system checks whether the resource is already occupied by another operation or whether it is locked during this period of time. A note is issued to show any possible conflicts. This requires that the option *Check for capacity availability* has been activated in the tab [planning component](#).



Specific licenses are required in order to check assignments and visualize occupied/locked resources.

Show material conflicts

If this option is enabled, the operation bar is displayed in the defined color in case of a material conflict.

Merged OPs

If you enable this option and you merge operations (OPs), the system displays a narrow bar in the configured color at the bottom of the bars of the respective merged operations in the Gantt chart.

Material available

If the option *Show material conflict* is enabled, the operation bar is displayed in the defined color if the required material is available.



To color the bars with reference to the material availability, the detail view *Planned inventory levels* must be open. The system performs the checks and colors the bars with reference to the material availability of the selected material.

Color of line marker

If you change the sequence of the operations in the workplace pool table, a line appears during the drag & drop action at the point where the operations are entered. You can configure the color of the line with the setting "Color of line marker".

HYDRA: as of service pack 13



FEDRA: as of version 1.1

You can display and color the line marker.

Highlight selected OP bars

Set the option "Highlight bar selection" and a frame is added to the highlighted OP bars. You can select the color and the thickness of the frame.

HYDRA: as of service pack 13



FEDRA: as of version 1.1

You can highlight selected OP bars.

Note available

If this option is set, the operation bar shows the symbol  indicating that a note is available for the operation.

Show maintenance



This option is only available if you enable the extension [graptmaint](#).

If this option is set, the system graphically notifies the user that a maintenance interval will end during processing of the planned operation. Further information on processing can be found in the section [Show maintenance](#).

Minimum bar length

If the remaining run time of an operation is less than or equal to 0, the processing time of this operation is shown with the duration entered in the field *Minimum bar length*. Consequently, you can still edit (select, move) the operation.

1.2.4 Tab *Tooltip/bar text*

In this tab, you can define the information that is displayed in the operation's tooltip or in its processing bar.

Tooltip

Click the section *Click here to add new tooltip* to select an additional text to be shown in the tooltip. A selection window opens. Click on the text you want to display in the tooltip. The selection window closes. The text is included in the list.

You can change the order in the tooltip. Click an entry in the list and move it without releasing the button. Release the left mouse button to drop it (drag and drop). The selected entry is moved behind the position where you release the left mouse button.

To delete an entry, select the entry you want to delete and click .

Available tooltip information

The information in the tooltip is read from the database when data is requested.

Some data might change dynamically during planning. Alternative production methods or the setup matrix can be named as examples. In the below table, this data is identified by an "X" in the column "dynamic modification".

Description	Dynamic modification
Order	
Sequence	
Operation	
Split	
MES order number (order/OP)	
Operation status	
Secondary status	
OP name	
Article	
Article designation	
Fixed (yes, no)	X
Note available (the internal indicator is shown): X - at least one note is available but no note is identified with the option "display on terminal". T - at least one note is available and at least one note is identified with the option "display on terminal".	
Tool	X
Farbe	
Material	
Target cycle	X
Actual cycle	
Target quantity (P) in the primary quantity unit	

Description	Dynamic modification
Yield (P) in the primary quantity unit	
Remaining quantity (P) (target quantity - yield in the primary quantity unit)	X
Unit (primary quantity unit)	
Partitioning	X
Setup time	X
Dynamic setup time (accounting for the setup change matrix)	X
Current setup time (Setup time + dynamic setup time)	X
Target duration (see note below!)	X
Earliest start	
Latest start	
Earliest end	
Latest end	
Scheduled start time	
Scheduled end time	
Planned start date	X
Planned end date	X
Basic start date	
Basic end date	
Priority	
Order index	
First logon	
Last logon	
User fields	

Notes on specific fields:

Target duration: This field does not display the net processing time according to the remaining run time formula. Instead it shows the bar length shown in the graphic planning board and does not account for shift breaks or any other breaks. The processing time depends on the status of the operation:

- For operations that are not logged on (status <> "running"), the bar length equals the sum total of the setup time + processing time according to the remaining run time formula + teardown time + dynamic setup time (if available);
- For operations that are logged on, the bar length equals the sum total of the processing time according to the remaining run time formula + retooling time.

The calculation is based on current values (e.g. remaining quantity to be produced). If a field that is included in the calculation of the remaining run time is changed in the database (e.g. because of an update in the PPS system), the changed value is only available once the planning board has been reloaded.

Tool: If you plan an operation by selecting a production variant for a workplace, the tool in the tooltip is no longer up-to-date. When you save the current planning, the correct tool (from the previously selected production variant) will be displayed in the tooltip.

User fields: You can select different user fields that either refer to operations or orders. The configuration of the user field keys AGNR/SYSTEM or AUNR/SYSTEM specifies the range of selectable user fields.



You must not assign a name to a user field that already exists in the standard (field, tooltip or bar text).

If you display the month in a user field, you must write the placeholders for the month in upper case letters in the type definition: xx.MM.xxxx.

Bar legend

Click the section *Click here to add new text* to select an additional text to be shown in the OP bar. A selection window opens. Click on the text you want to display in the OP bar. The selection window closes. The text is included in the list.

To delete an entry, select the entry you want to delete and click .

Available information

Description
Order
Sequence
Operation
Split
MES order number (order/OP)
Operation status
OP name
Article
Article designation
Tool
Target cycle
Priority
Order index



You need to refresh data in the planning board to make sure the configurations are enabled.

1.2.5 Workplace tab

Visible columns

The left hand side of the planning board shows the capacity groups including the assigned individual capacities (workplaces/machines) that match the planner's responsibility area or user profile.

You can display the following columns in this list of workplaces:

Column	Meaning
status	Status color or RPA color (depending on the option settings for "Use status text colors")
Workplace	Workplace/machine number
Short name	Short name of the workplace.
Designation	Full name of the workplace
Group	Group of the workplace.
Symbol	Graphic See also the description for the button "Download workplace icons from server".
Performance level	Current performance level of the workplace.

Change the column widths directly in the table (click and drag across). This modification is saved user-specifically.

We recommend to display at least the workplace/machine number.

Use status text colors

If this option is enabled, the workplaces are colored according to the currently active status. The individual status color is defined in the assigned status text. If no color is defined for a status text, the workplace is shown without a color (white/transparent).

If this option is not set, the machines are colored according to the configured RPA.

Show expected malfunction period

When the workplace/machine status is updated, the expected duration of a malfunction entered in the Windows terminal can also be transferred.

If this option is set and if a status is available for which the duration > 0, then the expected duration is made visible in the planning board. The expected end is calculated:

Expected end = maschinen_status.prog_begin_dat/zeit + maschinen_status.prog_dauer
(compared to the Gregorian calendar)

If the expected end is in the future, then the period of time between "now" and the expected end is shown as a red bar. The bar does also cover periods without shift (times without shift are not considered). At the same time, a respective symbol is displayed in the column *Symbol* of the workplace list. This symbol overrides the symbol that is normally displayed.

Tooltip

The red bar includes a tooltip showing the below-mentioned information:

- Workplace: workplace number according to configuration
- Short name: short name of the workplace according to configuration
- Designation: name of the workplace according to configuration
- Current status: status number and status text of the status currently available for the workplace.
- Status since: date, time
- Expected malfunction period: duration in hrs:min:sec.
- Expected end: date, time

Conversion of the malfunction period into times without shift

The red bar includes a context menu. This menu includes the menu item "Convert to time without shift". Once you have clicked this menu item, the dialog *Individual shift times* opens. The period of time starting from "now" until the expected end is pre-assigned.

Remove malfunction period

If the malfunction duration returns to 0 the next time the status is identified, the bar and the symbol are removed and the initial symbol is shown again. If this period has been defined as time without shift, this period remains and the planner must remove it explicitly.

Show staff assignment



This option is only available if the modification [grapt.spe](#) is enabled and the user is assigned the function authorization *grapt.spe*.

The personnel planned for a workplace in the [Workplace assignment](#) application is shown, if this option is enabled. You can find further information about how to display planned staff in the graphic planning [here](#).

Change the color of the workplace if an OP with production variant is selected

Set this option to show the workplaces in a specified color where an operation can be processed according to defined production variants. Select the color for the workplaces where an operation can be processed according to defined production variants.

If you select an operation with a defined production variant, all machines are displayed in color that can be used to produce the operation's article/item on the left hand side of the workplace/machine display area. Clicking an empty space in the planning board will remove this color marking.

Selection of production variants of manual planning

In the section *Selection of production variants of manual planning*, select the option *Automatic selection*, if only one production variant exists when you manually plan an operation on a workplace and you want to select the production variant automatically if only one production variant is stored for the article in the workplace.

Generate a production variant for machines from a production variant for groups

You can define production variants for groups if you do not specify a workplace/machine when creating the production variant. If you manually plan an operation in the planning board, you can plan this OP on every workplace of the group.

If the option *Generate a production variant for machines from a production variant for groups* is set and the planner has selected a group production variant during manual planning, the system prompts the planner to confirm that he/she would like to use this production variant to generate a production variant for machines. After confirming the prompt, the system generates a production variant for machines.

Download workplace icons from server

By default, the following symbols are displayed:

- Single workplace according to configuration: 
- Group workplace according to configuration: 

Using this button, the planner can download workplace-specific symbols from the server and store them locally so that they are displayed in the list of workplaces.

For this, the system selects and transfers all files following the scheme *_small.png in the directory defined via the logical [Path](#) MOCWPIMG.

When the list of workplaces is loaded, the system checks for every workplace if a file exists in the local directory. If a file exists, the system displays this file (this image). If there is no file, the system checks if the local directory includes a file for the group the workplace is assigned to. If there is a file for the group, this file is displayed.

The search process uses the file name configured for the workplace in the [Resource configuration](#). File sizes should be as small as possible for the display in the list of workplaces. For this reason the file must comply with the naming conventions described below:

Example: File name configured for the workplace: 12330.jpg

Extract the front part of the file name → 12330

Add "_small" → 12330_small

Add the file extension ".png" → 12330_small.png

Search for file 12330.png

If found, the image 12330_small.png is displayed in the list of workplaces.

Otherwise:

Display the default image depending on the workplace type as configured:

- E = Individual workplace:



- G = Group workplace:



1.2.6 Tab *Priorities*

Display priorities

You can use the priority of an operation as control tool. The priority is a single digit, numeric value. The value increases in ascending order ("0" = lowest priority, "9" = highest priority).

If this option is enabled, the operation's priority is shown on the left as a colored triangle:



You can define the color for the individual priorities as you like.

Also in the toolbar, you can activate/deactivate the symbol used to display the priority.

Show delays

Using this option, you can identify operations that are expected to be delayed. A narrow bar is displayed on top of the operation bar, if the planned end of a planned operation is x hours earlier than the latest end of the operation (LET). Coloring depends on the configured hours.

1.2.7 Tab "Operation links"

Function authorization: op.cmbbuild

Here you can undertake configurations that have an effect on the way the function works and on the visualization of joint productions. You can find further information at [This document](#).

1.2.8 Tab Views

Resource types

The selection of resource types in the drop-down menu controls the display of resources in the resource view. Use the drop-down menu to filter the resource types. Only for the selected resource types, the respective resources are then displayed. You must request data, to activate the filter.

You can find the icon to activate the resource view in the toolbar, tab *Planning details*.

1.3 Tabular *Pool of groups*

Go to tab [Planning details](#) in the toolbar and click the icon *Pool of groups*.

The tabular pool of groups shows the operations planned for a group. It shows the operations of *all* groups of the planning board. The planner can individually define the sorting of the table.

Using the table's filter function, you can restrict the pool of groups to single groups. Refer to the documents on the general operation ([moc_cc.pdf](#)) for detailed information on the use of the filter function.

User fields: You can select different user fields that either refer to operations or orders. The configuration of the user field keys AGNR/SYSTEM or AUNR/SYSTEM specifies the range of selectable user fields.

1.4 Tabular *Pool of workplaces*

The tabular pool of workplaces shows all planned operations of a workplace in a tabular list. The list shows operations sorted by their planned start date. However, running operations are always displayed on top of the list, chronologically sorted by logon time.

It is possible to replan operations using the pool of workplaces tables. If you drag an operation to the top of the list (so that the drop line is at the top), the operation is planned at the start of the planning horizon. If you drag an operation A so there is operation B above the drop line, the planned start of operation A is put at the end of operation B.

It is not possible to change the display with regard to the sorting of operations via the table sorting function. Also, you cannot use the grouping function, as a change in the display sequence affects the sequence of operation planning.

You can switch to another workplace's pool of workplaces using one of these two options:

- Calling the menu item in the workplace's context menu in the Gantt workplace table
- Single clicking on the required workplace in the Gantt workplace table

The fields that are available in the tabular pool of workplaces are the same as the ones in the [tabular pool of groups](#).

Single clicking an operation in the tabular pool of workplaces

If you use the mouse and single click on an operation with the left button, the system searches for the relevant operation in the other currently active views/dialogs and selects it.

Double clicking an operation in the tabular pool of workplaces

If you use the mouse and double click on an operation with the left button, the system opens the [Order overview](#) dialog. When doing so, the system also transfers the respective order/operation. In the order overview, the system automatically displays the order in the upper detail application and the order's operations in the lower detail application (order progress).

User fields: You can select different user fields that either refer to operations or orders. The configuration of the user field keys AGNR/SYSTEM or AUNR/SYSTEM specifies the range of selectable user fields.

1.5 Notes

You can enter notes on an operation. You can view the notes already entered for an operation using the detail application *Notes on operations*, which can be activated from the [Toolbar](#).

The detail application includes a table and a memo field. The table shows the following columns on the left hand side: short text, modified by, date, time, display on terminal, MES order number. To the right, the memo text is displayed. The position of the splitter between table and memo field is saved.

If the detail application *Notes* is active, an additional tab *Notes* is shown in the toolbar. This tab contains the functions needed to edit notes:

Insert a note

You can insert a new note using the operation's context menu. You require the function authorization edopnote.create to insert a new note. The dialog matches the [editing dialog](#) available in the order management. When you call the dialog, the MES order number of the operation is pre-assigned. You must enter a short text and a long text.

Disable the checkbox *Display at terminal* if you do *not* want the note to be displayed on the shop floor terminal (CTWIN, AIP).

Edit a note

You can edit an existing note with the corresponding buttons in the toolbar.

Delete a note

You can delete an existing note with the corresponding buttons in the toolbar.

Visualization with the operation bar

The icon  displayed with the operation bar of the graphic planning board (Gantt) indicates if at least one note is assigned to an operation. If the detail application *Notes* is visible and you click one or more OP bars, the contents of the detail application *Notes* are updated.

1.6 Conflict list



You can use the button  to call the conflict list from the graphic planning board. It shows the planning conflicts for the current planning profile. The conflict list is always updated online, i.e. it is always updated after each planning action.

In case of manual user activities, the conflict list is only updated when the user activity is fully completed. So, if any validation checks occur that must be edited by the planner, the conflict list is not updated. The conflict list is only updated when the activity has been fully completed.

In case of automatic planning activities, the conflict list is only updated when the entire planning activity has been fully completed.

Conflicts displayed

By default, the conflict list shows the conflicts listed below. You can use the combo box *Conflict* to restrict the displayed conflicts.

Conflicts that are only shown for planned OPs

- Violation of a relationship AOB
- Overloaded resource RES
- Planned end lies past latest end of the operation PEG
- Planned start lies ahead of earliest start of the operation PSG
- Planned start lies ahead of earliest start of the order PSA
- Planned end lies past latest end of the order PEA
- Material not available (planned inventory levels) MVPMAT
- Material not available (energy consumption) MVPEMG
- Insufficient staff available PEPKAPA
- Insufficient staff available with required qualification PEPQUALI

Conflicts shown for planned and unplanned OPs:

- Scheduled end lies past latest end of the operation	TEG
- Scheduled start lies ahead of earliest start of the operation	TSG
- Scheduled start lies ahead of earliest start of the order	TSA
- Scheduled end lies past latest end of the order	TEA
- Personnel not available (Person OK = "N")	PNV
- Tool not available (Tool OK = "N")	WNV
- Material not available (Material OK = "N")	MNV



The fields "Person OK", "Tool OK" and "Material OK" are filled, if you change the resource status (e.g. in the application "Order overview", function *Modify resource status*).

Field description

The following fields (columns) can be displayed in the conflict list.

Conflict

Text description of the conflict: see above "Displayed conflicts"

Order / sequence / OP / split / MES order number

Operation triggering a conflict.

OP name

Name of the operation according to the order backlog

Article

Article number of the operation

Article designation

Article designation

Resource

Conflict "Overloaded resource" (RES): The workplace/machine or the resource is overloaded because of the scheduling of the operation. Usually, this conflict is the result of a double assignment. This conflict might also be caused if you try to plan the operation during a period when the resource is locked.

Other conflict: Workplace/machine or resource the operation is planned for.

Group

Group of the workplace/machine.

Planned start

Planned start date (date, time) when the operation is planned.

Planned end

Planned end (date, time) of the operation. The planned end is calculated using the planned start plus the remaining run time. The result is compared to the (planning) shift model of the workplace and the performance level stored in the workplace.

Earliest start

Earliest start date (date, time) of the operation according to lead time scheduling.

Earliest end

Earliest end date (date, time) of the operation according to lead time scheduling.

Latest start

Latest start date (date, time) of the operation according to lead time scheduling.

Latest end

Latest end date (date, time) of the operation according to lead time scheduling.

Scheduled start time

Scheduled start (date, time) of the operation.

Scheduled end time

Scheduled end (date, time) of the operation.

Order start date

Basic start date (date, time) of the order.

Order finish date

Basic end date (date, time) of the order.

Actual start

For running operations: Point in time (date, time) since when the operation has been logged on. For operations that are processed at workplaces with activated shift automatic, the actual start is the point in time of the beginning of the shift, if a shift change has taken place since the operation has been logged on.

Conflict with OP

In case of a conflict of type "Violation of a relationship"(AOB), the MES order number of the operation is displayed here.

Because the "other" operation is also "involved" in this conflict, a conflict is displayed for this operation as well.

Information

If the following conflicts occur, this field shows the material number of the material component that will not be fully available at the planned point in time:

- "Material not available (planned inventory levels)" (MVPMAT),
- "Material not available (energy consumption)" (MVPEMG).

Priority

Priority of operation

Defect code

Internal number of the conflict

Running

Indicates if an operation is currently logged on (status "running")

Planned

Indicates if an operation is currently planned for a workplace.

Fixed

Indicates if the operation is fixed.

Target duration

The target duration is not the net processing time according to the remaining run time formula. Instead it refers to the bar length shown in the graphic planning board and does not account for shift breaks or any other breaks. The processing time depends on the status of the operation:

- For operations not logged on (status $\neq$ "running"), the bar length is the sum total of the setup time + dynamic setup time (if available) + processing time according to the remaining runtime formula + teardown time.
- For running operations, this is the sum total of the processing time according to the remaining run time formula + teardown time.

The calculation is based on current values (e.g. remaining quantity to be produced).

If a field included in the calculation of the remaining run time formula is changed in the database (e.g. because of an update from the ERP system), then this changed value will not be taken into account until the next time the planning board is reloaded.

Actual duration

Sum total of the times posted to the resource performance accounts 1 - 11.

1.7 Processing notes

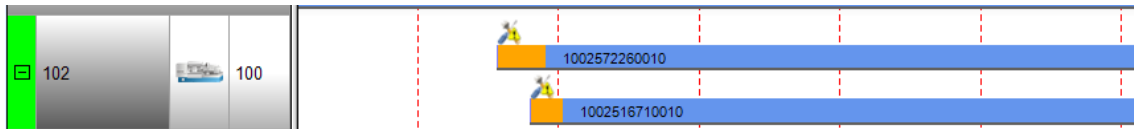
1.7.1 Show maintenance



The function described below is only available if the extension [graptmaint](#) is enabled.

The system can show if a maintenance becomes due while a planned operation is processed. You can configure due dates for maintenance activities in the MOC application [Maintenance calendar / Activity calendar](#). You must enable the option *Show maintenance* in the HLS settings, in tab [Bar layout](#), to have a maintenance displayed that becomes due.

The icon  above the operation bar indicates that during run time of the OP a maintenance is due for the workplace where the OP is planned or for the resource assigned to the OP:



The system displays the following maintenance types:

- Maintenance based on cycles
- Maintenance based on time
- Maintenance based on hours of operation

Maintenance dates are recalculated every time the graphic planning board is re-planned. The maintenance status "red" (compare configuration in the MOC application [Maintenance calendar / Activity calendar](#)) indicates when a maintenance activity is due.

The system calculates the next maintenance due dates based on the operations included in the current planning profile or planning horizon.

If maintenance is due for a running operation, the following is assumed when calculating the next maintenance due date:

- The maintenance activity is performed after finishing the operation
- After the maintenance activity has been completed, the system calculates the next maintenance dates for subsequent operations.

In case multiple operations are planned simultaneously and use the same resources, the due date for the next maintenance is calculated separately for each operation (setup times not included).



For technical reasons, maintenance due dates are not shown immediately after an automatic assignment but only once you have refreshed data.

Calculation with cycle-based maintenances

To calculate a maintenance due date in the future (resource cycles required by an operation), use the following formula and include the operations planned for the resource:

$$\text{Cycles} = (\text{target quantity} (\text{primary}) - \text{yield} (\text{primary})) / \text{partitioning of operation}$$

The system rounds up calculated cycles to integer values for each operation. If the calculated cycle value is less than 0, the value will be set to 0. The value "cycles recorded so far" defined for the maintenance is used as the initial value.

Tooltip display

The tooltip shows the affected resource and maintenance type. The tooltip also shows if several maintenance activities become due during the operation's planned term. These activities are automatically shown below the configured values. The following information is shown for a maintenance:

- Resource including resource type due for maintenance
- Type, class and description of due maintenance activity

For reasons of clarity, the tooltip shows a maximum of five maintenance activities at a time for an operation.